

* Series

المتسلسلات

Math 2

Date / / Subject

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots$$

For-ex $1 + 3 + 5 + 7 + \dots$ (arithmetic Series)

$2 + 4 + 8 + 16 + \dots$ (Geometric Series)

* Convergence & divergence of Series

التقارب والتباعد للمتسلسلات

$$\sum_{n=1}^{\infty} a_n = \begin{cases} = \infty & \text{divergent (متباعدة)} \\ < \infty & \text{convergent (متقاربة)} \end{cases}$$

* Ratio test اختبار النسبة

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \begin{cases} < 1 & \text{convergent} \\ > 1 & \text{divergent} \\ = 1 & \text{failed} \end{cases}$$

* Revision

$$\textcircled{1} \frac{a^m}{a^l} = a^{m-l}$$

$$\textcircled{2} a^m * a^l = a^{m+l}$$

$$\textcircled{3} n! = n * (n-1) * (n-2) * \dots * 3 * 2 * 1$$

factorial (مضروب)

$$5! = 5 * 4 * 3 * 2 * 1$$

$$4! = 4 * 3 * 2 * 1$$

$$5! = 5 * 4!$$

$$(n+1)! = (n+1) * n!$$

* Limits to infinity (∞)

$$\lim_{n \rightarrow \infty} f(n) = ?$$

الخطوات كالآتي:-

① نعوض تعويض مباشر مع ملاحظة الآتي

$$\textcircled{P} 0 = \frac{\text{عدد}}{\infty}$$

$$\textcircled{Q} \infty = \frac{\text{عدد}}{0}$$

يوجد احتمالين بعد التعويض

② الاحتمال الأول الناتج يكون رقم

③ الاحتمال الثاني الناتج يكون $\left[\frac{\infty}{\infty} \right]$ كمية غير معينة

نقسم البسط والمقام على أكبر أس موجود لـ (n)

$$\textcircled{EX} \lim_{n \rightarrow \infty} \frac{3n^5 + 3n}{n^5 + 4}$$

$$= \frac{3\infty + \infty}{\infty + 4} = \frac{\infty}{\infty}$$

بالقسمة بسطاً ومقاماً على (n^5)

$$\lim_{n \rightarrow \infty} \frac{3 + \frac{3}{n^4}}{1 + \frac{4}{n^5}} = \frac{3+0}{1+0} = \boxed{3}$$

EX) DISCUSS the convergence of Series

$\sum_{n=1}^{\infty} \frac{1}{n!}$ using Ratio test

$$a_n = \frac{1}{n!}, \quad a_{n+1} = \frac{1}{(n+1)!}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{(n+1)!}}{\frac{1}{n!}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{(n+1)!} \times \frac{n!}{1} = \lim_{n \rightarrow \infty} \frac{n!}{(n+1)!}$$

$$= \lim_{n \rightarrow \infty} \frac{n!}{(n+1) \times n!} = \lim_{n \rightarrow \infty} \frac{1}{(n+1)}$$

$$= \frac{1}{\infty+1} = \frac{1}{\infty} = 0 < 1$$

Series is convergent

EX) DISCUSS the convergence of Series

$\sum_{n=1}^{\infty} \frac{3^n}{2^n}$ using Ratio test

$$a_n = \frac{3^n}{2^n} \Rightarrow a_{n+1} = \frac{3^{(n+1)}}{2^{(n+1)}}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{3^{(n+1)}}{2^{(n+1)}} \times \frac{2^n}{3^n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{3^{(n+1)}}{3^n} \right) \times \left(\frac{2^n}{2^{(n+1)}} \right) = \lim_{n \rightarrow \infty} (3^{(n+1)-n} \times 2^{n-(n+1)})$$

$$= \lim_{n \rightarrow \infty} (3 \times 2^{-1}) = \lim_{n \rightarrow \infty} \left(\frac{3}{2} \right)$$

$$= \frac{3}{2} > 1$$

Series divergent

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EX) DISCUSS the convergence

$\sum_{n=1}^{\infty} \frac{2^n}{n!}$ using ratio test

$$a_n = \frac{2^n}{n!}, \quad a_{n+1} = \frac{2^{(n+1)}}{(n+1)!}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \left[\frac{2^{(n+1)}}{(n+1)!} \times \frac{n!}{2^n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{2^{(n+1)}}{2^n} \times \frac{n!}{(n+1)!} \right]$$

$$= \lim_{n \rightarrow \infty} \left[2^{(n+1)-n} \times \frac{n!}{(n+1) \times n!} \right]$$

$$= \lim_{n \rightarrow \infty} \left[\frac{2}{1} \times \frac{1}{(n+1)} \right] = \lim_{n \rightarrow \infty} \left[\frac{2}{n+1} \right]$$

$$= \frac{2}{\infty+1} = 0 < 1$$

Series is convergent

P-Series test

A Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ مقارنة مع $\frac{1}{n^p}$

- $\rightarrow P > 1$ convergent
- $\rightarrow P \leq 1$ divergent

EX) DISCUSS the convergence of Series

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ using P-Series test

compare with $\sum_{n=1}^{\infty} \frac{1}{n^p}$

$$P = 2 > 1$$

Series is convergent

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Ex) Discuss the convergence of Series

$$\sum_{n=1}^{\infty} \frac{3}{\sqrt{n}} \text{ using P-Series test}$$

Compare with $\sum_{n=1}^{\infty} \frac{1}{n^p}$

$$\frac{3}{\sqrt{n}} = \frac{3}{n^{\frac{1}{2}}}$$

$$p = \frac{1}{2} < 1 \Rightarrow \text{Series is divergent}$$

Ex) Discuss the convergence using P-Series test of

$$\sum_{n=1}^{\infty} \frac{5}{\sqrt[4]{n^3}}$$

compare with $\sum_{n=1}^{\infty} \frac{1}{n^p}$

$$\frac{5}{\sqrt[4]{n^3}} = \frac{5}{(n^3)^{\frac{1}{4}}} = \frac{5}{n^{\frac{3}{4}}}$$

$$p = \frac{3}{4} < 1 \Rightarrow \text{Series divergent}$$

Ex) Discuss the convergence of Series

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^4} \text{ using P-Series test}$$

compare with $\sum_{n=1}^{\infty} \frac{1}{n^p}$

$$\frac{\sqrt{n}}{n^4} = \frac{n^{\frac{1}{2}}}{n^4} = \frac{1}{n^4 \times n^{-\frac{1}{2}}} = \frac{1}{n^{\frac{7}{2}}}$$

$$p = \frac{7}{2} > 1 \Rightarrow \text{Series convergent}$$

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Ex) $\sum_{n=1}^{\infty} \frac{5^n}{5^n n^{1.1}}$ using P-Series

$$= \sum_{n=1}^{\infty} \frac{5^n}{5^n n^{1.1}} = \sum_{n=1}^{\infty} \frac{1}{n^{1.1}}$$

compare with $\sum_{n=1}^{\infty} \frac{1}{n^p}$

$$p = 1.1 > 1$$

Series is convergent